

## Chemistry: Periodic Trends

Atomic radius decreases across a period and increases down a group. Ionization energy increases across a period and decreases down a group. Electronegativity follows the same trend. Ionic bonds form between metals and non-metals; covalent bonds between non-metals. VSEPR theory predicts molecular geometry. Reaction rate =  $k[A]^m[B]^n$ ; activation energy determines reaction speed. Catalysts lower activation energy.